UNITED STATES OF AMERICA
BEFORE THE
FEDERAL ENERGY REGULATORY COMMISSION

Interconnection of Large Loads to the	)	
Interstate Transmission System	)	Docket No. RM26-4-000
	)	

TRANSMITTAL LETTER AND PUBLIC COMMENTS OF J. VANN CUNNINGHAM

June 23, 2026

To the Honorable Secretary:

Pursuant to the Commission's ongoing regulatory proceedings and the landmark June 18, 2026 orders directing regional transmission networks to reform large-load interconnection processes, I respectfully submit this Transmittal Letter and accompanying Technical White Paper for formal inclusion in the public record of Docket No. RM26-4-000.

The attached white paper, titled "Interconnection Governance: Stiff-Point Qualification Standards and Demand Caps as the Policy Response to Data Center Grid Stress," provides actionable, technically grounded frameworks designed to accelerate data center integration. Crucially, these frameworks prevent the externalization of localized network upgrade costs onto residential ratepayers and safeguard regional grid reliability.

As an IEDC Fellow with a fifty-year career spanning nuclear facility power siting, linear nodal network operations, and corporate industrial location, my analysis bridges grid engineering physics and macroeconomic development. Specifically, this paper outlines a path to deploy fault-level qualification standards, derived from the established IEEE 519-2022 standards and historical electric arc furnace interconnection precedents, alongside node-level demand caps to optimize existing grid infrastructure.

Respectfully submitted,
/s/ J. Vann Cunningham
J. Vann Cunningham, IEDC Fellow
Principal Director, Global Facility Location Consulting, Lockwood Greene Consulting (Ret.)
AVP Economic Development, BNSF Railway Co. (Ret.)
Assistant Director for Power Marketing, Tennessee Valley Authority (Ret.)
1309 Lyra Lane, Arlington, Texas 76013
B155549@gmail.com
817-371-9834

Enclosure: Technical White Paper and Comments of J. Vann Cunningham

UNITED STATES OF AMERICA
BEFORE THE
FEDERAL ENERGY REGULATORY COMMISSION

Interconnection of Large Loads to the	)	
Interstate Transmission System	)	Docket No. RM26-4-000
	)	

TECHNICAL WHITE PAPER AND COMMENTS OF J. VANN CUNNINGHAM, IEDC FELLOW, ON GRID STIFF-POINT CRITERIA AND DEMAND CAPS

Submitted by:

J. Vann Cunningham, IEDC Fellow
Principal Director, Global Facility Location Consulting, Lockwood Greene Consulting (Ret.)
AVP Economic Development, BNSF Railway Co. (Ret.)
Assistant Director for Power Marketing, Tennessee Valley Authority (Ret.)
1309 Lyra Lane, Arlington, Texas 76013 B155549@gmail.com 817-371-9834

TABLE OF CONTENTS

EXECUTIVE SUMMARY

The United States data center industry is approaching an inflection point that closely parallels the nuclear power retrenchment of the early 1980s. Demand forecasts projecting a near-doubling of data center power consumption by 2028 rest on assumptions that do not adequately account for the transition from training to inference workloads, for semiconductor-level efficiency improvements, or for the hard ceiling imposed by grid transmission capacity. A governance framework characterized by non-disclosure agreement opacity, cost externalization onto residential ratepayers, and an interconnection queue that rewards position over siting quality is generating community and regulatory resistance that has already resulted in an estimated $100 billion in delayed or canceled investment nationally.[1]

The administrative remedy is available, technically grounded, and does not require new legislation. Regional Transmission Organizations and Independent System Operators, operating under existing Federal Energy Regulatory Commission authority, administer the interconnection queue through which every large-load developer must pass. Extending the fault-level qualification standards that utilities applied to electric arc furnace operators beginning in the 1970s to data center interconnection, combined with node-level demand caps embedded in interconnection agreements, would implement a stiff-point siting standard that directs large digital infrastructure loads toward the transmission nodes capable of absorbing them without cascading grid stress or ratepayer cost socialization.

[1] Data Center Watch (10a Labs), *Q2 2025 Update* (April–June 2025), https://www.datacenterwatch.org/q22025. The report documents that at least 20 data center projects worth approximately $98 billion were blocked or delayed by local opposition in the second quarter of 2025 alone, a pace that continued to accelerate, reaching $156 billion for all of 2025 and approximately $130 billion in the first quarter of 2026 alone.

This paper develops that argument across five sections. Section I establishes the governance failure and its historical parallel. Section II presents the technical framework: stiff-point qualification, network optimization as the cost-effective alternative to new generation, and the arc furnace interconnection precedent. Section III applies the framework geographically across three distinct infrastructure tiers: the Northern Virginia exchange ecosystem, the Rust Belt industrial corridor, and the Southwestern markets. Section IV presents the RTO and ISO reform pathway. Section V presents consolidated policy recommendations.

The author brings direct practitioner experience to each dimension of this argument. Nuclear facility siting at TVA, manufacturing and industrial facility location across hundreds of projects including mini steel mills requiring arc furnace interconnection review, railroad economic development at BNSF, and international advisory work with the IMF, World Bank, and UNDP inform an analysis that integrates grid engineering, economic development practice, regulatory history, and infrastructure finance in a manner that disciplinary boundaries would otherwise prevent.

Summary of Policy Recommendations

1. FERC rulemaking establishing fault-level adequacy as a condition of large-load interconnection eligibility, with technical specifications based on IEEE standards.
2. RTO and ISO tariff amendments implementing node-level demand caps, set at available fault-level capacity net of existing committed load.
3. State-level or federally-baseline disclosure requirements mandating public documentation of projected resource consumption, emissions profiles, and employment commitments.

4. Priority queue treatment for interconnection applications at nodes with demonstrated stiff-point characteristics, particularly legacy heavy industrial sites in the Rust Belt corridor.
5. Long-term planning framework recognizing small modular reactor co-location as the firm power resolution for both Rust Belt distributed inference infrastructure and Southwestern closed-loop facility models.

SECTION I: THE GOVERNANCE FAILURE AND ITS HISTORICAL PARALLEL

The Nuclear Retrenchment as Governing Precedent

The nuclear retrenchment of the early 1980s was not caused primarily by the Three Mile Island incident, which was a technically containable event with limited radiological consequences. It was caused by the convergence of three conditions whose combination proved insurmountable: demand forecasts built on assumptions that had ceased to hold, a planning process insulated from public scrutiny by institutional opacity, and a regulatory framework that, once public legitimacy collapsed, had no mechanism to distinguish sound projects from unsound ones. All projects became politically vulnerable because the institutional framework for evaluating them had lost credibility.

The Tennessee Valley Authority's experience is instructive in the specific way this paper requires. TVA's multi-unit construction program was premised on a one-to-one relationship between economic growth and electrical demand. The energy crises of the 1970s broke that relationship permanently through structural decoupling: conservation and efficiency gains allowed industrial and commercial output to grow without proportional increases in kilowatt-hour consumption. The baseline forecasting assumption that had justified the entire construction program vanished. By 1982, the formal cancellation and suspension of the remaining units was the institutional acknowledgment that long-range planning models had been fundamentally upended by a changed economic reality.[2]

[2] Tennessee Valley Authority, TVA Annual Report and Strategic Power Program Review (Knoxville, TN: TVA, 1982). This institutional record documents TVA's formal cancellation or suspension of multiple nuclear construction projects, including Phipps Bend Units 1-2, Hartsville Units A1-A2 and B1-B2, and Yellow Creek Units 1-2, following the collapse of regional load-growth forecasts and the resulting reassessment of the Strategic Power Program.

The data center industry in 2026 exhibits all three conditions of the nuclear retrenchment in recognizable form. This is not a rhetorical analogy. It is a structural diagnosis.

The Demand Forecasting Problem

Current projections indicate that U.S. data center power demand will nearly double by 2028, rising from approximately eighty gigawatts to approximately one hundred fifty gigawatts, an increase of roughly seventy gigawatts.[3] This trajectory is consistent with a broader national pattern: five-year electricity load growth forecasts across strategic industries, including data centers, advanced manufacturing, and electrification, increased from approximately twenty-three gigawatts to approximately one hundred twenty-eight gigawatts between 2022 and 2024.[4]

Technology-driven resource demand historically follows a logistic growth curve: increasing at a decreasing rate until an equilibrium is reached, not the exponential or geometric trajectory that current projections imply. Several moderating forces are already visible. The data center boom has been driven primarily by the training phase of artificial intelligence, the extraordinarily power-intensive process of building and refining large foundational models. As the technology matures, the economics shift toward inference: running established models to serve user queries. Inference requires significantly less

---

[3] Bloom Energy, *2026 Data Center Power Report: When Power Defines Growth* (San Jose, CA: Bloom Energy, January 2026). The report projects that U.S. data center IT load capacity will rise from approximately 80 gigawatts in 2025 to approximately 150 gigawatts in 2028, an increase of roughly 70 gigawatts over three years.

[4] Grid Strategies LLC, Strategic Industries Surging: Driving U.S. Power Demand (National Load Growth Report, December 2024). The report finds that national five-year electricity load growth forecasts increased from approximately 23 gigawatts to approximately 128 gigawatts between 2022 and 2024, driven primarily by data centers, advanced manufacturing, and electrification.

energy per transaction than training. Simultaneously, semiconductor economics are producing neuromorphic architectures and algorithmic efficiency gains that execute complex computations on a fraction of the energy that earlier generations required.[5]

The structural risk is not that demand will collapse. It is that siting decisions made on peak-trajectory assumptions will be exposed by a demand moderation that arrives faster than the infrastructure committed to serve it.

The Transparency Deficit

A review of thirty-one Virginia municipalities with existing or proposed data centers found that approximately eighty percent had non-disclosure agreements in place with developers, limiting public access to information about project scale, resource needs, and potential impacts.[6]

This is a governance and regulatory failure, not simply a corporate communications failure. The NDA culture that produced eighty percent opacity in Virginia localities is not a legitimate business practice in the context of infrastructure that depends on public transmission systems. It is an information asymmetry that systematically prevents communities from quantifying and negotiating the costs they are being asked to absorb. The Environmental Impact Statement requirement under NEPA did not stop nuclear

[5] International Energy Agency, *Energy and AI* (Paris: IEA, April 2025). The report documents that AI inference now accounts for a substantial and growing majority of data center computing energy use relative to training, and that efficiency gains in accelerator hardware, model architecture, and algorithmic optimization have meaningfully reduced the energy required per computation even as aggregate demand continues to grow.

[6] The statistical metrics regarding municipal non-disclosure agreements cited herein are derived from the investigative findings published by Bonds, E., & Newby, V. (2025, April 30), "Data centers, non-disclosure agreements and democracy," Virginia Mercury. The author relies on the factual reporting of this established journalistic entity, as a matter of public record, and has not independently audited the proprietary FOIA tracking logs of the underlying 31 municipalities.

development. It imposed a transparency standard that forced utilities to defend their siting decisions against public scrutiny, separating the defensible projects from the indefensible ones. The data center industry requires an equivalent mechanism.

The Cost Externalization Pattern

The deepest structural problem is the systematic externalization of infrastructure costs onto parties who had no role in the siting decision. Residential ratepayers fund grid upgrades they did not request and from which they derive no direct benefit. Communities absorb impacts from industrial land use, diesel particulate emissions from backup generators, noise, and water consumption without the information necessary to negotiate mitigation. The interconnection queue, which nationally had grown to over two thousand gigawatts of pending requests by 2024, with PJM accounting for a substantial share of that backlog, functions on a first-come, first-served allocation mechanism that rewards queue position rather than siting quality.[7]

The consequence of this cost externalization, measured in concrete terms, is an estimated one hundred billion dollars in delayed or canceled projects nationally between mid-2025 and 2026. That figure is the measurable output of a planning model that generates systematic opposition wherever it is applied. The institutional constraint amplifies grid and resource constraints by converting manageable discrete delays into compounding schedule and cost failures.

[7] Center on Global Energy Policy at Columbia University, Outlook for Pending Generation in the PJM Interconnection Queue (2024). This analysis documents PJM's severe interconnection backlog and the resulting multi-year study freezes and restructuring of its interconnection process. U.S. Department of Energy, Transmission Interconnection Roadmap (2024). DOE reports that more than 2,600 gigawatts of clean-energy and storage projects were sitting in interconnection queues nationwide by 2024, of which PJM accounted for a substantial share.

The Jobs Gap as a Structural Grievance

Community resistance to data center development is not irrational opposition to infrastructure. It reflects a reasonable assessment of the cost-benefit distribution. A hyperscale facility drawing hundreds of megawatts typically employs twenty-five to one hundred fifty permanent workers. The nuclear plants, steel mills, and manufacturing operations that anchor comparative expectations employed thousands. Communities are being asked to absorb the cost profile of heavy industrial infrastructure while receiving a fraction of the employment return. This paper's geographic recommendations for Rust Belt brownfield development and its long-range framework for small modular reactor co-location address this structural grievance directly.

SECTION II: THE TECHNICAL FRAMEWORK

Stiff-Point Grid Siting: Definition and Rationale

A stiff point, in transmission network engineering, is a node on the grid where fault-level capacity is sufficient to absorb large, variable industrial loads without propagating voltage instability, harmonic distortion, or power quality degradation through the surrounding network. The concept is not theoretical. It is an established engineering parameter embedded in large industrial interconnection protocols administered by utilities and regional grid authorities for decades.

Data centers are among the most power-demanding and operationally sensitive industrial loads in the modern economy. A hyperscale facility drawing two hundred megawatts requires continuous, firm power at voltage and frequency standards that computing equipment can tolerate within narrow deviation bands, with redundancy sufficient to maintain operations through grid stress events. This demand profile is not meaningfully different, from a network engineering perspective, from the demand profile of an electric arc furnace installation. Both impose large, variable, power-quality-sensitive loads on the transmission network. Both require adequate fault-level capacity at the interconnection node to contain the disturbance locally rather than allowing it to cascade through interconnected systems.

The critical difference is that the electric arc furnace interconnection standard required demonstration of fault-level adequacy and installation of reactive power compensation equipment as conditions of interconnection approval. No equivalent standard currently governs data center interconnection. The result is a systematic mismatch between the locations where data center demand is concentrated and the locations where the grid can actually absorb it.

The Arc Furnace Interconnection Precedent

The arc furnace interconnection standard is the direct technical and regulatory precedent for the framework this paper proposes. When mini steel mills deploying electric arc furnaces began proliferating in the 1970s and 1980s, utilities and regional grid operators confronted a specific engineering problem: arc furnaces produce severe voltage fluctuations and harmonic distortions that can propagate across interconnected networks if the host node lacks sufficient fault-level strength to contain them. The response was to condition interconnection approval on two requirements. The prospective customer had to demonstrate that the proposed site carried sufficient fault-level capacity to absorb the load. The prospective customer also had to install reactive power compensation equipment, including static VAR compensators and related apparatus, to contain power quality disturbances within the local interconnection boundary.

This framework was applied across hundreds of mini steel mill siting decisions throughout the industrial expansion of that period. It did not stop mini steel mill development. It shaped where that development went and who bore the cost of network adequacy. The author sited a substantial number of mini steel mill facilities during this period, working through this interconnection qualification process from the developer's side. That direct experience with the arc furnace standard is not incidental to this analysis. It is the practical foundation for the claim that stiff-point qualification, as an interconnection condition, is administratively workable, technically defensible, and does not suppress development of facilities at genuinely qualified sites.

Network Optimization Outperforms New Generation

When evaluating how to resolve a regional infrastructure deficit created by large-load additions, the comparison between new baseload generation and transmission network optimization reveals a clear practical hierarchy. Bringing new baseload capacity online requires multi-year regulatory processes, large upfront capital commitments with long amortization periods, and supply chain dependencies that further extend timelines.

Increasing the stiffness of an existing transmission node operates on a fundamentally different timescale and at a fraction of the capital cost. The relevant upgrades, replacing aging switchgear, installing static synchronous compensators and reactive power compensation equipment, and reconductoring transmission lines within existing rights-of-way, occur within established utility operating footprints under existing regulatory frameworks. The requirement that large-load interconnection applicants demonstrate node adequacy, or fund the specific localized upgrades required to achieve it, places network optimization costs on the party whose siting decision creates them. It also creates a market signal that favors sites where the foundational engineering is already in place.

Demand Caps: Mechanism and Rationale

A demand cap, as used here, is a ceiling on the aggregate load that can be attached to a specific interconnection node or substation before the regional system requires material investment to maintain stability and power quality. Caps of this kind are administratively familiar: utilities set analogous limits on the aggregate load served from individual distribution feeders, and RTOs impose comparable constraints on the aggregate generation that can interconnect at specific transmission nodes. The extension of this logic to large-load interconnection is analytically straightforward.

The demand cap mechanism directly addresses the cost externalization problem. Under the current framework, a developer who saturates a node's available fault-level capacity imposes upgrade costs on the regional system, which are then socialized across all ratepayers. A demand cap that halts new interconnection approvals at a node once its available capacity has been committed forces the next developer to either identify a different node, fund the upgrade required to expand capacity at the chosen node, or wait until existing load is reduced. In each case, the cost of the siting decision is borne by the party making it.

The combination of stiff-point qualification and demand caps produces a geographic distribution of large-load development that aligns naturally with grid topology: high fault-level nodes receive queue priority, while nodes lacking adequate fault-level capacity are effectively disqualified unless the developer funds the necessary upgrades. This converts grid physics into market signals without prescriptive land use regulation.

Anticipated Objection: Barrier to Entry and Incumbent Protection

A predictable objection to the stiff-point qualification standard is that it functions as a barrier to entry protecting incumbent operators in established high-capacity clusters, particularly Northern Virginia, from competition by making it structurally difficult to develop at nodes in markets where grid investment has historically been deferred. The objection deserves direct engagement because it is technically sophisticated and will be raised by parties with genuine standing in any comment proceeding.

The objection fails on both the historical record and the logical structure of the proposed framework. The arc furnace interconnection standard did not protect incumbent steel producers. It shaped where new entrants sited their facilities by directing them toward nodes with adequate fault-level capacity. The standard was competitively neutral: it applied equally to all prospective customers regardless of their relationship to existing operators. New mini steel mill development proceeded efficiently at qualified sites throughout the industrial expansion of the 1970s and 1980s.

The logical structure of the stiff-point framework produces the same result. Northern Virginia carries high fault-level capacity at its major data center interconnection nodes precisely because decades of utility investment have built infrastructure to serve concentrated industrial load. A developer proposing a new facility at a Loudoun County node faces the same stiff-point qualification requirement as a developer proposing a facility at a legacy steel corridor node in Ohio. If the Loudoun County node has available fault-level capacity net of existing committed load, the qualification is met. If it does not, because existing load has saturated node capacity, the demand cap operates as intended: it directs the next increment of development to a node that can absorb it without ratepayer cost socialization.

The framework is pro-competitive in a specific and important sense: it creates market conditions under which new development geographies, the Rust Belt corridor in particular, can compete for investment on the merits of their grid topology. The priority queue treatment proposed for legacy heavy industrial sites in Recommendation 4 makes this pro-competitive intent explicit. Far from protecting incumbents, the stiff-point standard is the mechanism through which the current incumbent concentration in Northern Virginia is most likely to be redistributed toward markets that the existing governance framework has not been able to reach.

SECTION III: GEOGRAPHIC APPLICATION ACROSS THREE INFRASTRUCTURE TIERS

The stiff-point framework is not a uniform national standard that applies identically across all geographies. The transmission network is not uniform, and the data center workload is not monolithic. Three distinct infrastructure tiers have emerged from the development history of the industry, each with distinct grid characteristics, workload profiles, and structural vulnerabilities.

Tier One: The Northern Virginia Exchange Ecosystem

Northern Virginia's dominance in data center development began with MAE-East, one of the first major Internet Exchange Points in the United States, established in the Ashburn submarket in the mid-1990s. The agglomeration dynamic that followed is technically precise: latency-sensitive applications cannot be relocated away from the exchange hub without accepting a measurable performance penalty, and each operator's presence makes the hub more valuable to the next. Dominion Energy invested in ultra-high-capacity transmission infrastructure tailored to the data center load profile concentrating in its service territory, creating utility infrastructure unavailable elsewhere.

A parallel and non-negotiable demand source reinforced this concentration. The federal intelligence and defense infrastructure surrounding Washington created a category of data center investment with no geographic flexibility. Classified and sensitive compartmented information workloads cannot be relocated based on land economics or power costs. The workforce dimension compounds this lock-in: Northern Virginia has the largest concentration of security-cleared personnel in the United States.

The structural lock-in of Northern Virginia is now generating the constraints that are redistributing investment. New facilities face power hookup delays. Land prices have risen to levels that compress development economics. The cluster has approached the carrying capacity of its local infrastructure. The stiff-point framework's application to this tier does not seek to redirect demand that genuinely requires co-location with the exchange ecosystem or carries federal security requirements. It governs the category of demand created by AI inference deployment at regional scale, which is geographically distributable and does not require co-location with Ashburn.

Tier Two: The Rust Belt Industrial Corridor

Legacy Rust Belt industrial districts carry precisely the grid characteristics that the stiff-point framework rewards. The industrial districts that supported heavy manufacturing, integrated steel production, and automotive assembly during the postwar decades required, and received, high-capacity transmission lines and substations engineered to absorb the kind of volatile, nonlinear loads that are exceptionally difficult to manage on a regional network. Electric arc furnaces, in particular, required fault-level capacity sufficient to contain their characteristic voltage fluctuations and harmonic distortions locally. That engineering legacy persists even where the industrial activity that justified it has long since ceased. A substation built to serve a now-idle steel facility still carries the fault-level capacity it was designed for.

The permitting advantages compound the grid argument. Legacy industrial zones are already zoned for heavy industrial use. Brownfield redevelopment under the established CERCLA framework operates within a defined regulatory structure rather than the open-ended NEPA review that governs greenfield development. Rights-of-way are already platted and legally established. Data centers can be sited on capped industrial slabs where the soil must remain undisturbed, converting long-standing environmental liabilities into revenue-generating digital assets.

As the AI industry transitions from training, which requires tightly coupled GPU clusters that resist geographic distribution, toward inference, the operational requirements shift in ways that favor the Rust Belt model. Inference workloads require substantially less power per transaction, benefit from proximity to end users rather than to other data centers, and tolerate geographic distribution across regional markets. The Rust Belt corridor's dense network of mid-sized metropolitan areas delivers sub-fifteen-millisecond latency to major Midwestern and Eastern population centers, well within the threshold for commercial enterprise AI execution. The investment flows confirm the transition is underway: Columbus and New Albany in Ohio have attracted multi-billion-dollar campus commitments from Amazon, Google, and Meta.

Tier Three: The Southwestern Markets

Texas and Arizona were chosen deliberately based on deregulated power economics, abundant land, renewable energy procurement advantages, and favorable tax regimes. Those advantages were real during the build-out phase. The Southwestern model also carried structural vulnerabilities that were underpriced at the time of the initial investment decisions.

Texas operates its primary electrical grid, ERCOT, as an island with minimal external interconnection. February 2021 demonstrated the consequence: a sustained cold-weather event simultaneously increased heating demand and caused widespread generator failures, producing rotating outages that affected millions of customers and caused deaths and economic losses estimated in the hundreds of billions of dollars. Data center operators responded with aggressive on-site generation and battery storage investment, which addresses the symptom rather than the structural cause.

Arizona built data center capacity on evaporative cooling in a desert supplied by the Colorado River via the Central Arizona Project. Lake Mead has been at historically low levels for most of the past decade. The federal government has imposed mandatory cutbacks on Colorado River deliveries, and municipalities in the Phoenix metropolitan area have restricted or conditioned new data center water hookups.[8]

The stiff-point framework's application to the Southwestern tier addresses a pattern that the Texas and Arizona experience illustrates: siting frameworks built on single-variable optimization tend to produce attendant vulnerabilities. Texas optimized for power costs without adequately pricing grid-isolation risk. Arizona optimized for cooling efficiency without adequately pricing water constraint. The long-term resolution for the Southwestern market is addressed in Section V's discussion of small modular reactors.

[8] Gerlak, A. K., et al., Scenario Planning: Embracing the Potential for Extreme Events in the Colorado River Basin, Climatic Change, 165(1), 27 (2021). U.S. Bureau of Reclamation, Colorado River Basin Lower Basin Shortage Guidelines and Water Management Alternatives (2023). This federal record outlines the tier-level shortage determinations for Arizona and the Lower Basin, including restrictions that have led metropolitan water authorities in the Phoenix region to condition, limit, or defer high-volume industrial cooling hookups for data-center development.

SECTION IV: THE RTO AND ISO REFORM PATHWAY

FERC's Existing Authority

The Federal Energy Regulatory Commission has existing authority under the Federal Power Act to require that interconnection studies account for power quality impacts, voltage stability, and fault-level adequacy at proposed interconnection nodes. FERC has used that authority in the generator interconnection context to impose technical requirements that effectively qualify and disqualify sites on engineering grounds. Extending that framework to large-load interconnection, with explicit stiff-point qualification criteria analogous to the fault-level standards that utilities applied to arc furnace operators, would implement the core siting argument through the regulatory mechanism that data center developers must engage regardless of their preferences.

RTOs and ISOs, operating under FERC oversight, administer the interconnection queue through tariff provisions that FERC approves. Tariff amendments implementing node-level demand caps can be implemented through the standard tariff filing process with stakeholder comment opportunities. This is the same mechanism through which FERC has historically implemented major interconnection reforms, including Order 888, Order 2000, and Order 2023. The institutional pathway is established and does not require new legislation.

The Queue Reform Dimension

As noted previously, the national interconnection queue, which grew to over 2,000 gigawatts of pending requests by 2024, operates on a first-come, first-served allocation mechanism that rewards queue position rather than siting quality. The vast majority of these requests will never result in operating facilities. They represent speculative queue positions held by developers who have not yet secured financing, permitting, or confirmed load commitments. The cost of processing these applications and the delay they impose on legitimate projects with confirmed load and sound siting is borne by the entire system.

FERC's Order 2023 addressed queue congestion by imposing more rigorous readiness requirements, including financial deposits and milestone demonstrations to filter speculative applications.[9]

Queue readiness reform and siting quality reform are complementary instruments. A developer with confirmed financing and committed load can still secure a queue position at a node with inadequate fault-level capacity, imposing upgrade costs on the system that a stiff-point qualification requirement would have prevented. A stiff-point qualification standard implemented as a condition for initiating an interconnection study would filter the queue at the point of entry, reducing study costs and shortening processing times for qualified applications.

The ERCOT Jurisdictional Boundary

The stiff-point qualification framework proposed in this paper operates within FERC's jurisdictional reach: PJM, MISO, SPP, WECC, and the other RTOs and ISOs subject to the Commission's oversight under the Federal Power Act. ERCOT, which operates as an intrastate system with minimal external interconnection, falls outside that

[9] Improvements to Generator Interconnection Procedures and Agreements, Order No. 2023, 184 FERC para. 61,054 (2023), order on reh'g and clarification, Order No. 2023-A, 186 FERC para. 61,199 (2024). These companion orders establish the Commission's first-ready, first-served cluster-study framework and impose heightened readiness and financial-security requirements on interconnection customers.

jurisdictional boundary by design. The Commission is aware of this structural limitation and this paper does not ask the Commission to act beyond its statutory authority.

The ERCOT boundary is nonetheless analytically relevant to this submission for a specific reason. The structural vulnerabilities documented in Section III, particularly the grid isolation that prevented import of surplus capacity during the February 2021 weather event and the water constraint affecting evaporative cooling systems in the Phoenix metropolitan area, illustrate what the governance framework proposed here is designed to prevent within FERC's jurisdiction. They are cautionary evidence, not a proposed remedial target. The Commission may wish to note, in any order or inquiry that results from this proceeding, that the data center industry's concentration in the ERCOT footprint represents a grid reliability exposure that falls outside federal oversight and that Congress may wish to consider in the context of broader energy infrastructure legislation. That observation is offered for the Commission's consideration rather than as a request for Commission action.

The Transparency Requirement as a Complement

The interconnection reform argument does not stand alone as a governance remedy. The NDA opacity documented across Virginia localities operates at the local level, outside FERC's direct jurisdiction, and cannot be resolved through interconnection queue reform alone. The stiff-point and demand cap framework addresses cost externalization through the transmission system. It does not address the information asymmetry that prevents communities from evaluating what a proposed facility will demand from local water systems, generate in diesel particulate emissions during backup generator activations, or contribute in permanent employment relative to the infrastructure burden it imposes.

A transparency requirement for large data center interconnection applications, mandating public disclosure of projected demand, water consumption, backup generation capacity and emissions profile, and expected permanent employment as part of the interconnection study process, would serve the function that NEPA's EIS requirement served in the nuclear era. It would not prevent sound projects from proceeding. It would prevent the information asymmetry that currently allows developers to externalize community costs while suppressing the information that would allow communities to quantify and negotiate those costs.

Several states have begun moving toward mandatory disclosure in response to community resistance and the documented NDA opacity. A federal baseline standard, whether through FERC guidance or conditions attached to transmission system access, would prevent the race-to-the-bottom dynamic in which developers migrate to the least transparent jurisdictions while the underlying grid stress follows them.

SECTION V: POLICY RECOMMENDATIONS AND LONG-RANGE FRAMEWORK

Near-Term Recommendations

Recommendation 1: FERC Rulemaking on Fault-Level Qualification

FERC should initiate a rulemaking establishing fault-level adequacy as a condition of large-load interconnection eligibility, with technical specifications derived from IEEE standards long applied by utilities to large industrial load interconnections.[10]

The rulemaking should define a minimum fault-level capacity threshold for data center interconnection applications above a specified megawatt threshold, require preliminary fault-level screening prior to initiation of full interconnection studies, and provide a pathway for developers to proceed at disqualified nodes by committing to fund required upgrades as a condition of interconnection. This rulemaking does not require new legislation. FERC's authority over transmission access and interconnection terms under the Federal Power Act is sufficient, and the agency has direct precedent for imposing technical qualification requirements on interconnection applicants in the generator interconnection context.

Recommendation 2: RTO and ISO Tariff Amendments for Node-Level Demand Caps

PJM, MISO, SPP, and other RTOs and ISOs operating under FERC jurisdiction should file tariff amendments implementing node-level demand caps for large-load interconnection. Caps should be set at the fault-level capacity available at each interconnection point, net of existing committed load.

[10] IEEE, IEEE Standard for Harmonic Control in Electric Power Systems, IEEE Std 519-2022 (Piscataway, NJ: Institute of Electrical and Electronics Engineers, 2022). This standard establishes the technical basis for fault-level and stiff-point qualification by defining allowable voltage-distortion limits, current-harmonic injection thresholds, and system-performance criteria for large nonlinear industrial loads.

The amendments should require developers who seek to attach load at a node whose available capacity has been committed to either demonstrate that proposed load falls within remaining capacity, fund upgrades required to expand node capacity in advance, or identify an alternative qualifying node. These amendments should be developed through the standard stakeholder process, providing the comment opportunity that also serves the transparency function the governance analysis identifies as necessary.

Recommendation 3: Mandatory Disclosure for Large-Load Interconnection Applications

FERC should establish, either through rulemaking or as a condition of RTO and ISO tariff approval, a mandatory disclosure requirement for large-load interconnection applications above a specified megawatt threshold. Required disclosures should include projected peak and average power demand over a ten-year facility horizon, projected water consumption by cooling technology type, backup generation capacity and type with associated emissions profiles under modeled activation scenarios, and expected permanent employment at the facility.

Disclosures should be filed as part of the public interconnection study record and made available to affected municipalities and state public utility commissions. This requirement does not prevent development. It prevents the information asymmetry that generates the community opposition that has produced one hundred billion dollars in project delays.

Recommendation 4: Priority Queue Treatment for Stiff-Point Qualified Sites

RTOs and ISOs should establish a priority processing track for large-load interconnection applications at nodes with demonstrated stiff-point characteristics. Legacy heavy industrial sites in the Rust Belt corridor that carried industrial loads of comparable or greater magnitude within the past thirty years should be eligible for expedited preliminary qualification based on documented interconnection history.

This recommendation addresses both the queue congestion problem and the geographic distribution objective simultaneously: it moves the highest-quality applications through the queue most efficiently while creating a market signal that favors development at sites the network can actually support.

Long-Range Framework: Small Modular Reactors and the Firm Power Resolution

The near-term recommendations address the governance and cost externalization problems through the interconnection framework. They do not address the firm power supply problem that the transition away from fossil fuel baseload generation creates for an infrastructure category that requires continuous, dispatchable power around the clock, in all seasons, regardless of weather or grid conditions.

Intermittent renewable generation cannot meet the firm power requirement directly. Solar generation is unavailable at night and reduced under cloud cover. Wind generation varies with atmospheric conditions on timescales that cannot be predicted or controlled. Battery storage addresses short-duration backup at acceptable cost for a single cycle, but the lifecycle cost incorporating two to three replacement cycles over a thirty-year facility life materially narrows the economic advantage over alternative firm power sources. Natural gas backup reintroduces carbon emissions and fuel supply chain exposure that large technology companies have publicly committed to eliminating from their operational footprints.

Small modular reactor technology, with units targeting fifty to three hundred megawatts of output, factory-fabricated modular components, passive safety systems, and siting footprints compatible with brownfield industrial parcels, is the most technically coherent long-range resolution to the firm power problem. Several SMR designs are in active NRC licensing review. Data centers need firm baseload power at the scale that SMR units deliver, in geographies where grid capacity is constrained, on a timeline that matches the ten-to-fifteen-year infrastructure planning horizon that serious data center operators apply to long-term strategic capacity requirements.

Recommendation 5: Federal SMR Co-Location Planning Framework

The Department of Energy, in coordination with FERC and the NRC, should develop a planning framework specifically addressing SMR co-location with data center infrastructure across two distinct deployment models.

In the Rust Belt corridor, the proximate co-location model, where an SMR sited on an adjacent industrial parcel connects to one or more data center campuses through a short dedicated transmission segment, represents the most commercially practical near-term configuration. The stiff-point transmission infrastructure at legacy heavy industrial sites provides grid connection in both directions: as a load demand center and as a generation source. Water availability in the Ohio River and Great Lakes watersheds is a baseline condition rather than an engineering challenge. The brownfield industrial land use history provides the separation from residential density that the NRC siting review process values.

In the Southwestern markets, the closed-loop infrastructure model, integrating on-site SMR generation with waste-heat absorption cooling and zero-liquid-discharge water remediation within the facility boundary, converts the extractive demand pattern of the initial build-out into a self-contained resource loop that addresses the grid isolation and water constraint problems simultaneously. Public power utilities in the TVA region and the Nebraska public power system, with their legally binding economic development mandates and long-term power purchase agreement financing structures, represent the most appropriate institutional partners for early SMR co-location deployment.

CONCLUSION

The data center industry is not facing a technology problem. The computing hardware is capable, the fiber topology is established, and the economic demand for digital infrastructure is real and growing. The constraint is governance, and the governance remedy is available, administratively grounded, and ready for use.

The nuclear retrenchment that opened this paper's historical analysis ended when the institutional framework governing the technology lost public legitimacy faster than it could be reformed. The technology itself survived, migrating offshore and into deferred domestic capacity that took four decades to reconsider. The data center industry is approaching the same inflection point. The estimated one hundred billion dollars in delayed and canceled projects is not a temporary political friction that improved public relations will resolve. It is the measurable output of a governance model that systematically externalizes costs, suppresses information, and treats the regulatory signal of community resistance as a public relations problem rather than an infrastructure planning failure.

Stiff-point qualification standards and node-level demand caps, implemented through the RTO and ISO interconnection framework that every large-load developer must engage, address the core governance failure: they direct development toward the geographies the network can support, impose the costs of unsuitable siting on the parties making those decisions, and provide the transparent technical record that communities and regulators need to evaluate proposed infrastructure on its actual merits. Mandatory disclosure requirements extend the transparency standard to the community impact dimensions that the interconnection framework alone does not reach.

The geographic application of this framework produces a distribution of data center development that the market is already moving toward independently, as Northern Virginia's structural constraints push inference workloads toward the Rust Belt industrial corridor and the Southwestern markets face the consequences of underpriced resource dependencies. The stiff-point framework does not redirect the market. It converts the direction the market is already moving into enforceable policy that protects ratepayers and communities in the process.

The long-range firm power resolution, through small modular reactor co-location at Rust Belt brownfield industrial sites and within the closed-loop infrastructure model in the Southwest, addresses the employment gap that data centers alone cannot close and the carbon-free baseload requirement that renewable generation with battery backup has not yet demonstrated at the scale and duration AI inference infrastructure requires. It is a ten-to-fifteen-year proposition, not a near-term fix. The near-term recommendations in this paper create the governance conditions under which that longer-range development becomes economically rational and politically legitimate.

The data center challenge is not technological. The constraint is governance, and the governance remedy is available.

ABOUT THE AUTHOR

J. Vann Cunningham is an IEDC Fellow with a fifty-year career spanning nuclear facility siting, manufacturing and industrial site selection, railroad economic development, and international advisory work.

At the Tennessee Valley Authority, Cunningham served as Chief of Regional Planning and Assistant Director of Industrial Development for Power Marketing. His work at TVA included direct involvement in nuclear facility siting during the 1970s construction program, providing the practitioner foundation for this paper's analysis of the nuclear retrenchment precedent and the NRC siting framework. During this period he also developed the SES Differentials Index at Oak Ridge National Laboratory, a theory-driven predictive framework for social conflict around energy facility siting grounded in Weber's class-status concepts and Smelser's structural strain theory, and the Bundle of Goods residential location model derived from analysis of over thirty-six thousand workers across seven TVA nuclear construction sites, which received the American Planning Association's student research award in 1981.

At Lockwood Greene, Cunningham served as Global Director of Economic Development and Corporate Facility Location. In this role, he directed hundreds of manufacturing and industrial facility-location engagements, including mini-steel mill siting projects requiring arc-furnace interconnection qualification. This direct experience with the stiff-point interconnection process from both the utility and developer sides is the practical foundation for the arc furnace precedent argument developed in Section II of this paper.

At BNSF Railway Company, Cunningham served as Assistant Vice President of Economic Development (retired), directing the railroad's economic development program across its network. Rail and electrical transmission share the fundamental structure of linear nodal networks, and the economic development work at BNSF reinforced the analytical framework applied here to data center siting and grid governance.

Cunningham's international advisory work has included engagements with the United Nations Development Programme on infrastructure development and economic development planning. His academic foundation is undergraduate social anthropology with minors in geology and epistemology, and graduate study in urban and regional planning at the University of Tennessee, Knoxville.

REFERENCES

Bloom Energy. *2026 Data Center Power Report: When Power Defines Growth*. San Jose, CA: Bloom Energy, January 2026.

Bonds, E., & Newby, V. Data centers, non-disclosure agreements and democracy. Virginia Mercury (Apr. 30, 2025).

Center on Global Energy Policy at Columbia University. Outlook for Pending Generation in the PJM Interconnection Queue (2024).

Data Center Watch (10a Labs). *Q2 2025 Update*. April–June 2025. https://www.datacenterwatch.org/q22025.

Gerlak, A. K., et al. Scenario Planning: Embracing the Potential for Extreme Events in the Colorado River Basin. Climatic Change, 165(1), 27 (2021).

Grid Strategies LLC. Strategic Industries Surging: Driving U.S. Power Demand (National Load Growth Report, 2024).

IEEE. IEEE Standard for Harmonic Control in Electric Power Systems, IEEE Std 519-2022. Piscataway, NJ: Institute of Electrical and Electronics Engineers, 2022.

Improvements to Generator Interconnection Procedures and Agreements, Order No. 2023, 184 FERC para. 61,054 (2023), order on reh'g and clarification, Order No. 2023-A, 186 FERC para. 61,199 (2024).

International Energy Agency. *Energy and AI*. Paris: IEA, April 2025.

Tennessee Valley Authority. TVA Annual Report and Strategic Power Program Review. Knoxville, TN: TVA, 1982.

U.S. Bureau of Reclamation. Colorado River Basin Lower Basin Shortage Guidelines and Water Management Alternatives (2023).

U.S. Department of Energy. Transmission Interconnection Roadmap (2024).